Ana Aquiles

PHD

KEY COMPETENCES

- Python Programing
- Scientific writing
- Scientific Illustration
- Computational Neuroscience
- Science Communication
- Calcium imaging in vitro
- Animal surgery / Stereotaxic surgery
- Miniscope recordings
- Image Analysis
- Experimental design
- Time-series Analysis
- Machine learning
- Deep Learning
- Open Science
- Rodent models
- Large dataset management
- Non-human EEG datasets
- MRI analysis

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RESEARCH POSITIONS

CNRS, Institut du Génomique Fonctionnelle — **Apr 2024 - Mar 2026**
Postdoctoral Fellow
Director: Patrice Mollard

CNRS, Institut du Génomique Fonctionnelle — **Sep 2023 - Apr 2024**
Research Data Engineer
Director: Patrice Mollard

UNAM, Cellular Physiology Neurosciences Institute, Mexico — **Jan 2023 - August 2023**
Ph.D candidate, Visiting.
Data analysis of electrophysiological data with population dynamics applications, machine learning / deep learning and information theory, at Cognitive Neuroscience Lab lead by
Dr. Román Rossi Pool.

UNAM, Neurobiology Institute, Juriquilla Qro. Mexico — **Aug 2019 - Dec 2023**
Ph.D. Student
Thesis : "Excitability evaluation in a cortical dysplasia rodent model."
Supervisor : Luis Concha

UNAM, Neurobiology Institute, Juriquilla Qro. Mexico — **Jan 2018 - August 2019**
Research Assistant
Research Assistant at Brain Connectivity laboratory, lead by MD.PhD Luis Concha, in chargue of the experimental and research design of one of the main projects that actually runs in the lab.

Accomplishments:

- Experimental design in rodent model.
- Data analysis and acquisition.
- Teaching post-graduate students about the same project.

Supervisor : Luis Concha

UNAM, Faculty of Sciences, Mexico City — **Aug 2016 - Jan 2018**
BS. in Biology. Graduate Student

Thesis : "Characterization of the cellular network of lactotrophs and thyrotrophs during the reproductive cycle in a rodent model."

Supervisor : Tatiana Fiordelisio Coll

EDUCATION

PhD in Biomedical Sciences
Neurobiology Institute, UNAM CAMPUS, Juriquilla. Queretaro | 08/2019 - 12/2023
Bachelor of Science in Biology
Faculty of Sciences, UNAM, Mexico | 08/2014 - 01/2018

SCHOLARSHIPS & GRANTS

FRM (Fondation pour la Recherche Médical) - Postdoctoral Researcher — **Apr 2023 - Mar-2026**
Institut de Génomique Fonctionnelle, CNRS, Montpellier France

GroPro travel award (poster) - Stirling UK — **Aug 2024**

CONACyT Graduate Scholarship, — **Aug 2019- Dec 2023**
(The Mexican National Council for Science and Technology), which fully supported my doctoral studies.

Scholarship to LASCON Computational Neuroscience SCHOOL, USP, Brazil — **Jan 2020**
Computational Neuroscience School at Instituto de Matemática e Estatística da USP. São Paulo, Brazil.

Scholarship for International Research Internship, UNAM — **Dec 2017 - Feb 2018**
Internship at Computational Neuroscience lab, Valparaíso University, Chile.
Internship at Computational Biology Lab, Fundación Ciencia y Vida, Santiago de Chile.

FORMATIONS AND SKILLS

Machine learning applied to image analysis - BIOCAMPUS — **Jun 2025**
Postdoctoral Fellow

ib Cegos. Advanced Python - CNRS Formation — **May 2025**
Postdoctoral Fellow

NEUBIAS workshop in bioimage analysis at Pasteur Institute, France — **May 2025**
Postdoctoral Fellow
Computational methods and application into segmentation and image treatment using Python

Advanced Specialization Training in Surgery (Rodents and Lagomorphs) - CNRS Formation — **May 2024**
Postdoctoral Fellow

Training in the Design and Implementation of Projects and Procedures - CNRS — **Mar 2024**
Postdoctoral Fellow

Neuromatch Computational Neuroscience school, online — **Jul 2020**
Ph.D student.

LASCON (Latin American School on Computational Neuroscience), Brazil — **Jan 2020**
Ph.D student.
Computational neuroscience methods and application to real data and models.

UV, University of Valparaiso. Fundacion Ciencia y Vida. Chile — **Dec 2016- Feb 2017**
BS. in Biology, Internship
Data Analysis and neurocomputation at Computational Neuroscience Lab lead by Dr. Patricio Orio

PEER-REVIEWED PUBLICATIONS

- **Aquiles, A.**, Fiordelisio, T., Luna-Munguia, H., & Concha, L. (2023). Altered functional connectivity and network excitability in a model of cortical dysplasia. *Scientific reports*, 13(1), 12335.
- Villaseñor, P. J., Cortés-Servín, D., Pérez-Moriel, A., **Aquiles, A**., Luna-Munguía, H., Ramirez-Manzanares, A., ... & Concha, L. (2023). Multi-tensor diffusion abnormalities of gray matter in an animal model of cortical dysplasia. *Frontiers in Neurology*, 14, 1124282.
- Santiago-Andres Y, **Aquiles A**, Taniguchi-Ponciano K, Salame L, Guinto G, Mercado M, Fiordelisio T. Association between Intracellular Calcium Signaling and Tumor Recurrence in Human Non-Functioning Pituitary Adenomas. *Int J Mol Sci.* (2024) Apr 3;25(7):3968. doi: 10.3390/ijms25073968. PMID: 38612778; PMCID: PMC11011867.
- **Aquiles A.** & Mollard. P. (2025). Uncovering Functional Balance in Pituitary Baselines States. *Under Revision in Journal of Neuroendocrinology*.
- **Aquiles, A**., Pinedo, L., Regalado, M., Luna-Munguia, H., Peña-Ortega, F., Concha, L., & Rossi-Pool, R. (2026). Assessing the degree of cortical dislamination through electrical pattern analysis. *iScience*, 29(6).
- **Aquiles, A.,** Aparicio Arias, J., Lafont, C., Hodson, D., Santiago-Andres, Y., Mollard, P., & Fiordelisio, T. (2026). Population geometry reveals directed coupling and transient bistability in spontaneous pituitary secretion. bioRxiv, 2026-04. *Under Revision in Nature Communications Physics.*
- **Aquiles, A**. (2026) The mosaic cortex: somatic mutation as a hidden dimension of neocortical evolution. *Submitted to Biological Theory.*

SUPERVISION AND MENTORING

Olimpia Ortega-Fimbres, MSc Student Insitute of Neurobiology, Quéretaro	**2022-2023**
Daniela Bravo-Coutiño, Bc Student Insitute of Neurobiology, Quéretaro	**2022-2023**
Paulina Villaseñor de la Barrera, MSc Student Insitute of Neurobiology, Quéretaro	**2021-2023**
David Cortés-Servin, MSc Student Insitute of Neurobiology, Quéretaro	**2020-2021**

REVIEWER EXPERIENCE

- Frontiers in Computational Neuroscience (Present)
- Frontiers in Integrative Neuroscience (Present)
- Biomarker Research (Present)

SCIENTIFIC COMMUNICATION

- Workshop for kids at spring festival in Mexico city (2018) - "I want to be an insect". Main: Communicate the importance of insects in polinzation and daily food production. **A. Aquiles** AND E. Choreño Parra.
- Workshop for teachers for the care center of children, ISSSTE in Mexico City (2020) - "Neurobiology of lies". **A.Aquiles**, A.López-Castro, A. Díaz, D. Munguía. Main: Communicate the neural basis of lies and the evolution of them in the early human life and their repercussions in social interactions.
- Writing, structure and direction of clip "Savage brains" in Brain week at Neurobiology Institute, UNAM, diffused on FB Neurobiology Institute channel (2020). **A.Aquile**s, A. López-Castro, I. Montiel, D. Munguía, D. Saavedra. Main: Communicate the diversity of complex structure and function adaptation of brains from different species.
- Writing, and strcuture podcast episodes for science communication tansmited via Spotify Podcast NEUROKUNI.
- Plenary talk in TALENT-LAND, Guadalajara, México, (2019) - "How to detect science fake-news in a technology era"- A.Aquiles AND A.Valdéz

ANCILLIARY ACTIVITIES

- Co-Founder of science communication students group at Neurobiology Institute **Neurokuni 2021-2022**
- Representative of graduate students in the gender equality group at the Institute of Neurobiology - **2022-2023**
- **Mentor Career at Neuromatch Computational Neuroscience School 2026**

LANGUAGES

Spanish - native **English**- proficient **French**- proficient **Portuguese** - intermediate